

APEXPLANET DATA ANALYTICS INTERNSHIP

FINAL REPORT

Executive Summary

As part of the ApexPlanet Data Analytics Internship, I completed multiple hands-on tasks involving data cleaning, SQL analysis, dashboard creation, statistical analysis, customer segmentation, and predictive modeling using the Superstore dataset.

Task Overview

Task 1 - Data Cleaning

- Cleaned and prepared the dataset for analysis.
- Handled missing values and checked data quality.

Task 2 - SQL Analysis

- Performed SQL queries to analyze customer, sales, and product data.
- Generated insights using filtering, grouping, and aggregation functions.

Task 3 - Dashboard Development

- Created visual dashboards to analyze sales and profit trends.
- Used charts and KPIs for business reporting.

Task 4 - Advanced Analytics

- Performed statistical analysis.
- Conducted hypothesis testing (T-Test).
- Applied K-Means clustering for customer segmentation.
- Built a Linear Regression model for sales prediction.

Key Insights

1. Technology products generated the highest sales.
2. Certain product categories contributed more profit than others.
3. Customers could be segmented into different groups based on purchasing behavior.
4. Statistical analysis revealed meaningful business patterns.
5. Predictive modeling helped estimate future sales trends.

Business Recommendations

1. Focus marketing efforts on high-performing product categories.
2. Improve strategies for low-profit products.
3. Target customer segments with personalized offers.

Tools Used

- Python
- Pandas
- NumPy
- SQL
- Tableau
- Matplotlib
- Seaborn
- Scikit-learn

Conclusion

This internship provided practical experience in data analytics, visualization, SQL, statistics, and machine learning. The project improved my ability to derive insights from data and present findings effectively.